

Proof by contradiction

In this type of proof, we assume that opposite of what we are trying to prove and get a logical contradiction.

Steps to be followed:

- (1) Assume that P is false.
- (2) Using this assumption, show a contradiction.

e.g.:- Show that $\sqrt{2}$ is an irrational number.

Sol.:- Let P : $\sqrt{2}$ is an irrational number.

Assume $\neg P$ is true i.e. $\sqrt{2}$ is a rational number.

$$\text{Let } \sqrt{2} = \frac{p}{q}$$

such that p and q have no common factor.

$$\Rightarrow \sqrt{2} q = p$$

$$\Rightarrow 2 q^2 = p^2$$

$$\Rightarrow p^2 \text{ is an even number.}$$

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[$\because$ if p^2 is even then p must be even]

$$\Rightarrow p = 2k \quad \forall k \in \mathbb{I}$$

$$\Rightarrow p^2 = 4k^2$$

$$\Rightarrow q^2 = \frac{p^2}{2} = 2k^2$$

$$\Rightarrow q^2 \text{ is also an even number.}$$

$\therefore 2$ is a common factor of p and q .

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which is a contradiction as p and q have not common factor..

Therefore, $\neg P$ is false and so P is True.
and $\sqrt{2}$ is an irrational number.

e.g:- Prove that there is no largest integer that is a multiple of 5 using proof by contradiction.

Sol:- Let P : There is no largest integer that is a multiple of 5.

We assume $\neg P$ is true i.e. there is a largest integer that is multiple of 5.

Let m is the largest integer that is multiple of 5.

$$\therefore m = 5k \text{ for some } k \in \mathbb{Z}$$

Now, consider the integer $m+5$.

$$m+5 = 5k+5 = 5(k+1)$$

This shows that $m+5$ is also a multiple of 5 and is greater than m as well.

And hence, this is a contradiction that m is the largest integer that is a multiple of 5, so our assumption is not true.

Hence, there is no largest integer that is a multiple of 5.

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To prove the conditional statement $p \rightarrow q$.
We assume both p and $\neg q$ are true.

Then consider $\neg q$ as a premise, we draw the conclusion $\neg p$.

Thus, we get the contradiction $p \wedge \neg p$.

Therefore, we say that our initial assumption is not true as $\neg q$ is false as p is assumed to be true.

Finally, $\neg q$ is false implies that q is true and hence $p \rightarrow q$.

Steps to be followed:

- (1) Assume both p and $\neg q$ are true.
- (2) Use $\neg q$ and show that p is false, which is a contradiction.

e.g.:- Prove the statement:

"If $3n+1$ is even, then n is odd" by using the method of proof by contradiction.

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Solⁿ :- Let $p: 3n+1$ is even.
 $q: n$ is odd.

We shall assume that p and $\neg q$ are true.
 $\therefore$ Let $3n+1$ is even and n is even.

So, $n = 2k$ where $k \in \mathbb{Z}$.

then,
$$3n+1 = 3(2k)+1$$
$$= 6k+1$$

Now, $\because$
$$6k+1 = 2(3k)+1$$

$\Rightarrow 6k+1$ is an odd number.

$\Rightarrow 3n+1$ is an odd number.

So, this is a contradiction to the assumption that $3n+1$ is even.

Hence, n is not even.

This proves the statement, "if $3n+1$ is even, then n is odd".

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eg:- Prove that the sum of two consecutive integers is odd.

sol:- Let a and b be two integers.

Here, P : a and b are two consecutive integers.

Q : $a+b$ is odd.

$\&$ We shall assume that P and $\neg Q$ are true.

Thus, a and b are consecutive integers and $a+b$ is even.

$\therefore a = k$ and $b = k+1$, $k \in \mathbb{Z}$.

Thus $a+b = k+k+1$
 $= 2k+1$ which is an odd integer.

Thus, this is the contradiction to the assumption that $a+b$ is even.

Hence, $a+b$ is odd.

This proves the statement, "the sum of two consecutive integers is odd".

Proof by Mathematical induction :-

Steps to be followed :

- (1) Show true for the first term mostly $n=1$. $\rightarrow$ Base Case
- (2) Assume true for $n=k$. $\rightarrow$ Induction hypothesis
- (3) Show true for $n=k+1$. $\rightarrow$ Conclusion.
- (4) Re state $\rightarrow \therefore$ By the process of mathematical induction given statement.....

e.g.:- Prove that :

$$3 + 6 + 9 + 12 + \dots + 3n = \frac{3n(n+1)}{2}$$

Proof:- Show true for $n=1$.

$$3(1) = \frac{3(1)(1+1)}{2}$$

$$\Rightarrow 3 = \frac{3(2)}{2}$$

$$\Rightarrow 3 = 3$$

Assume true for $n=k$

$$3 + 6 + 9 + 12 + \dots + 3k = \frac{3k(k+1)}{2}$$

Show true for $n=k+1$

$$3 + 6 + 9 + 12 + \dots + 3k + 3(k+1) = \frac{3(k+1)(k+1+1)}{2}$$

$$\frac{3k(k+1)}{2} + 3(k+1) = \frac{3(k+1)(k+2)}{2}$$

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$$\Rightarrow \frac{3K(K+1) + 6(K+1)}{2} = \frac{3(K+1)(K+2)}{2}$$

$$\Rightarrow \frac{3(K+1)(K+2)}{2} = \frac{3(K+1)(K+2)}{2}$$

$$\Rightarrow \text{L.H.S.} = \text{R.H.S.}$$

Hence, by the process of mathematical induction

$$3+6+9+12+\dots+3n = \frac{3n(n+1)}{2}$$

Try this one yourself

* Using mathematical induction, prove that for every natural number.

$$1+2+3+\dots+n = \frac{n(n+1)}{2}$$